

Kessence`particles`hyperbolized`tanh`to`evol`test`mu

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Fields

$Psisf, phi, Phid_x, Phid_y, Phid_z, Psi$

Spatial Coordinates

x, y, z

Time Coordinate

t

Parameters

Parameter	Type	Default value
<i>tend</i>	REAL	30
<i>max_error</i>	REAL	$1e-10$
<i>phi_asymptotic</i>	REAL	0
<i>Phid_x_asymptotic</i>	REAL	0
<i>Phid_y_asymptotic</i>	REAL	0
<i>Phid_z_asymptotic</i>	REAL	0
<i>Psisf_asymptotic</i>	REAL	0
<i>phi_falloff</i>	REAL	1
<i>Phid_x_falloff</i>	REAL	2
<i>Phid_y_falloff</i>	REAL	2
<i>Phid_z_falloff</i>	REAL	2
<i>Psisf_falloff</i>	REAL	1
<i>Aphio</i>	REAL	Not set
<i>Aphi1</i>	REAL	Not set
<i>sigmax</i>	REAL	Not set
<i>betax</i>	REAL	Not set
<i>gammax</i>	REAL	$-1e17$
<i>Mpl_cte</i>	REAL	0.1994711401
<i>p_sigma</i>	REAL	3.35
<i>phi_init</i>	REAL	0.0
<i>p_Omega</i>	REAL	Not set
<i>tini</i>	REAL	0.0
<i>tslow</i>	REAL	Not set
<i>torbit</i>	REAL	800.0
<i>tslow2</i>	REAL	2.0
<i>sigmaslow</i>	REAL	10.0
<i>mass</i>	REAL	1.4
<i>massfactor</i>	REAL	2.167815001
<i>xcenter1</i>	REAL	0
<i>ycenter1</i>	REAL	0 2
<i>zcenter1</i>	REAL	0
<i>rorbit</i>	REAL	15.0
<i>rstart</i>	REAL	150.0
<i>rdonut</i>	REAL	4.0
<i>Nstar</i>	INT	1

Auxiliary Fields

phi2, dphi2

Auxiliary Variables

Xphi, Kf, KXf, KXXf, Phiu_x, Phiu_y, Phiu_z, Tbar, r1_x, r1_y, r1_z, sq_r1, r2_x, r2_y, r2_z, sq_r2, omegaorbit, rorbit_dynamic

Analysis Fields

phi22, vxplus, vxminus, vyplus, vyminus, vzplus, vzminus, vorticity_norm, dztphi, Chi_t, Chi_x, Chi_y, Chi_z, Chi_norm

Auxiliary Analysis Variables

Xphi, Kf, KXf, KXXf, Phiu_x, Phiu_y, Phiu_z, Tbar, r1_x, r1_y, r1_z, sq_r1, r2_x, r2_y, r2_z, sq_r2, omegaorbit, rorbit_dynamic, Vtx, Vty, Vtz, Vxy, Vxz, Vyz

Imported Models

K-essence flat

Region

main
Interior models
K-essence flat

Spatial Domain

Coordinate	Min	Max
<i>x</i>	-4	4
<i>y</i>	-4	4
<i>z</i>	-4	4

Initial Conditions

The condition is applied when:

$$Nstar = 1 \quad (1)$$

The condition is:

$$pos_x = x - xcenter1 \quad (2)$$

$$pos_y = y - ycenter1 \quad (3)$$

$$pos_z = z - zcenter1 \quad (4)$$

$$r = \max\{\sqrt{pos_x \ pos_x + pos_y \ pos_y + pos_z \ pos_z}, 0.0001\} \quad (5)$$

$$phi = 0 \quad (6)$$

$$\text{readFromFileQuintic}(1, 1, r, phi, 0) \quad (7)$$

$$drphi = 0 \quad (8)$$

$$\text{readFromFileQuintic}(1, 1, r, drphi, 0) \quad (9)$$

$$Phid_z = \frac{pos_z}{r} \ drphi \quad (10)$$

$$Phid_x = \frac{pos_x}{r} \ drphi \quad (11)$$

$$Phid_y = \frac{pos_y}{r} \ drphi \quad (12)$$

$$Psi = 0.0 \quad (13)$$

$$Psisf = nu \ phi \quad (14)$$

$$phi2 = phi \ phi \quad (15)$$

$$dphi2 = (Phid_x \ Phid_x + Phid_y \ Phid_y + Phid_z \ Phid_z) \quad (16)$$

The condition is applied when:

$$Nstar = 2 \quad (17)$$

The condition is:

$$pos_x = x \quad (18)$$

$$pos_y = y \quad (19)$$

$$pos_z = z \quad (20)$$

$$r = \max\{\sqrt{pos_x \ pos_x + pos_y \ pos_y + pos_z \ pos_z}, 0.0001\} \quad (21)$$

$$phi = 0 \quad (22)$$

$$\text{readFromFileQuintic}(1, 1, r, phi, 0) \quad (23)$$

$$drphi = 0 \quad (24)$$

$$\text{readFromFileQuintic}(1, 1, r, drphi, 0) \quad (25)$$

$$Phid_z = \frac{pos_z}{r} \ drphi \quad (26)$$

$$Phid_x = \frac{pos_x}{r} \ drphi \quad (27)$$

$$Phid_y = \frac{pos_y}{r} \ drphi \quad (28)$$

$$Psisf = nu \ phi \quad (29)$$

$$Psi = 0.0 \quad (30)$$

$$phi2 = phi \ phi \quad (31)$$

$$dphi2 = (Phid_x \ Phid_x + Phid_y \ Phid_y + Phid_z \ Phid_z) \quad (32)$$

Segment interactions

Interaction 1

Target segments: x-Lower, x-Upper, y-Lower, y-Upper, z-Lower, z-Upper

interaction

Analysis Field Equations

Analysis field equation

$$phi_{22} = Op(x, y, z, t) \tag{33}$$

$$\begin{aligned}
Op(x, y, z, t) = & -\frac{1}{4.0 \text{ Mpl_cte}} \frac{Tbar}{KXf - 2 KXXf Psisf^2} \\
& + \frac{KXf}{KXf - 2 KXXf Psisf^2} \partial_x Phid_x \\
& + \frac{KXf}{KXf - 2 KXXf Psisf^2} \partial_y Phid_y \\
& + \frac{KXf}{KXf - 2 KXXf Psisf^2} \partial_z Phid_z \\
& + (-4) KXXf Psisf \frac{Phid_x}{KXf - 2 KXXf Psisf^2} \partial_x Psisf \\
& + \frac{2 KXXf Phid_x^2}{KXf - 2 KXXf Psisf^2} \partial_x Phid_x \\
& + (-4) KXXf Psisf \frac{Phid_y}{KXf - 2 KXXf Psisf^2} \partial_y Psisf \\
& + \frac{4 KXXf Phid_x Phid_y}{KXf - 2 KXXf Psisf^2} \partial_x Phid_y \\
& + \frac{2 KXXf Phid_y^2}{KXf - 2 KXXf Psisf^2} \partial_y Phid_y \\
& + (-4) KXXf Psisf \frac{Phid_z}{KXf - 2 KXXf Psisf^2} \partial_z Psisf \\
& + \frac{4 KXXf Phid_x Phid_z}{KXf - 2 KXXf Psisf^2} \partial_x Phid_z \\
& + \frac{4 KXXf Phid_y Phid_z}{KXf - 2 KXXf Psisf^2} \partial_y Phid_z \\
& + \frac{2 KXXf Phid_z^2}{KXf - 2 KXXf Psisf^2} \partial_z Phid_z
\end{aligned} \tag{34}$$

vxplus

$$vxplus = (x, y, z, t) \tag{35}$$

$$\begin{aligned}
(x, y, z, t) = & + \frac{1.0}{KXf - 2.0 Psisf^2 KXXf} \left((-2.0 Psisf Phid_x KXXf) \right. \\
& \left. + \sqrt{KXf (KXf + 2 Xphi KXXf)} \right)
\end{aligned} \tag{36}$$

vxminus

$$vxminus = (x, y, z, t) \quad (37)$$

$$(x, y, z, t) = + \frac{1.0}{KXf - 2.0 \text{ Psisf}^2 KXXf} \left((-2.0 \text{ Psisf Phid}_x KXXf) \right. \\ \left. - \sqrt{KXf (KXf + 2 Xphi KXXf)} \right) \quad (38)$$

vyplus

$$vyplus = (x, y, z, t) \quad (39)$$

$$(x, y, z, t) = + \frac{1.0}{KXf - 2.0 \text{ Psisf}^2 KXXf} \left((-2.0 \text{ Psisf Phid}_y KXXf) \right. \\ \left. + \sqrt{KXf (KXf + 2 Xphi KXXf)} \right) \quad (40)$$

vyminus

$$vyminus = (x, y, z, t) \quad (41)$$

$$(x, y, z, t) = + \frac{1.0}{KXf - 2.0 \text{ Psisf}^2 KXXf} \left((-2.0 \text{ Psisf Phid}_y KXXf) \right. \\ \left. - \sqrt{KXf (KXf + 2 Xphi KXXf)} \right) \quad (42)$$

vzplus

$$vzplus = (x, y, z, t) \quad (43)$$

$$(x, y, z, t) = + \frac{1.0}{KXf - 2.0 \text{ } Psisf^2 \text{ } KXXf} \left((-2.0 \text{ } Psisf \text{ } Phid_z \text{ } KXXf) + \sqrt{KXf \text{ } (KXf + 2 \text{ } Xphi \text{ } KXXf)} \right) \quad (44)$$

vzminus

$$vzminus = (x, y, z, t) \quad (45)$$

$$(x, y, z, t) = + \frac{1.0}{KXf - 2.0 \text{ } Psisf^2 \text{ } KXXf} \left((-2.0 \text{ } Psisf \text{ } Phid_z \text{ } KXXf) - \sqrt{KXf \text{ } (KXf + 2 \text{ } Xphi \text{ } KXXf)} \right) \quad (46)$$

Analysis field equation

$$vorticity_norm = Op(x, y, z, t) \quad (47)$$

$$Op(x, y, z, t) = + ((-2) \text{ } Vtx^2 - 2 \text{ } Vty^2) - 2 \text{ } Vtz^2 + 2 \text{ } Vxy^2 + 2 \text{ } Vxz^2 + 2 \text{ } Vyz^2 \quad (48)$$

Analysis field equation

$$dztphi = Op(x, y, z, t) \quad (49)$$

$$\begin{aligned}
Op(x, y, z, t) = & -\frac{2}{2 KXXf Psisf^2 - KXf} KXXf Phid_x^2 \partial_x Phid_x \\
& -\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_y Phid_x \partial_y Phid_x \\
& -\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_z Phid_x \partial_z Phid_x \\
& +\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_x Psisf \partial_x Psisf \\
& -\frac{2}{2 KXXf Psisf^2 - KXf} KXXf Phid_y^2 \partial_y Phid_y \\
& -\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_z Phid_y \partial_z Phid_y \\
& +\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_y Psisf \partial_y Psisf \\
& -\frac{2}{2 KXXf Psisf^2 - KXf} KXXf Phid_z^2 \partial_z Phid_z \\
& +\frac{4}{2 KXXf Psisf^2 - KXf} KXXf Phid_z Psisf \partial_z Psisf \\
& -\frac{1}{2 KXXf Psisf^2 - KXf} KXf \partial_z Phid_z \\
& -\frac{1}{2 KXXf Psisf^2 - KXf} KXf \partial_y Phid_y \\
& -\frac{1}{2 KXXf Psisf^2 - KXf} KXf \partial_x Phid_x \\
& +\frac{\frac{1}{Mpl_cte}}{2 KXXf Psisf^2 - KXf} \frac{Tbar}{4}
\end{aligned} \tag{50}$$

Analysis field equation

$$Chi_t = Op(x, y, z, t) \tag{51}$$

$$Op(x, y, z, t) = +KXf Psisf \tag{52}$$

Analysis field equation

$$Chi_x = Op(x, y, z, t) \tag{53}$$

$$Op(x, y, z, t) = +KXf Phid_x \tag{54}$$

Analysis field equation

$$Chi_y = Op(x, y, z, t) \quad (55)$$

$$Op(x, y, z, t) = +KXf \ Phid_y \quad (56)$$

Analysis field equation

$$Chi_z = Op(x, y, z, t) \quad (57)$$

$$Op(x, y, z, t) = +KXf \ Phid_z \quad (58)$$

Analysis field equation

$$Chi_norm = Op(x, y, z, t) \quad (59)$$

$$Op(x, y, z, t) = +(-Chi_t^2) + Chi_x^2 + Chi_y^2 + Chi_z^2 \quad (60)$$

Auxiliary Analysis Equations

Auxiliary analysis variable equation

$$Phiu_x = Op(x, y, z, t) \quad (61)$$

$$Op(x, y, z, t) = +Phid_x \quad (62)$$

Auxiliary analysis variable equation

$$Phiu_y = Op(x, y, z, t) \quad (63)$$

$$Op(x, y, z, t) = +Phid_y \quad (64)$$

Auxiliary analysis variable equation

$$Phiu_z = Op(x, y, z, t) \quad (65)$$

$$Op(x, y, z, t) = +Phid_z \quad (66)$$

Auxiliary analysis variable equation

$$Xphi = Op(x, y, z, t) \quad (67)$$

$$Op(x, y, z, t) = +(-Psisf^2) + Phiu_x Phid_x + Phiu_y Phid_y + Phiu_z Phid_z \quad (68)$$

Auxiliary analysis variable equation

$$Kf = Op(x, y, z, t) \quad (69)$$

$$Op(x, y, z, t) = +0.5 Xphi ((-sigma_x) + 0.3333333333333333 gamma_x Xphi^2) + 0.25 beta_x Xphi^2 \quad (70)$$

Auxiliary analysis variable equation

$$KXf = Op(x, y, z, t) \quad (71)$$

$$Op(x, y, z, t) = +(-0.5 sigma_x) + 0.5 beta_x Xphi + 0.5 gamma_x Xphi^2 \quad (72)$$

Auxiliary analysis variable equation

$$KXXf = Op(x, y, z, t) \quad (73)$$

$$Op(x, y, z, t) = +0.5 beta_x + gamma_x Xphi \quad (74)$$

Auxiliary analysis variable equation

$$r1_x = Op(x, y, z, t) \quad (75)$$

$$Op(x, y, z, t) = +x + 2 \frac{mu}{1 + mu} rorbit_dynamic \cos (omegaorbit\ t) \quad (76)$$

Auxiliary analysis variable equation

$$r1_y = Op(x, y, z, t) \quad (77)$$

$$Op(x, y, z, t) = +y + 2 \frac{mu}{1 + mu} rorbit_dynamic \sin (omegaorbit\ t) \quad (78)$$

Auxiliary analysis variable equation

$$r1_z = Op(x, y, z, t) \quad (79)$$

$$Op(x, y, z, t) = +z \quad (80)$$

Auxiliary analysis variable equation

$$sq_r1 = Op(x, y, z, t) \quad (81)$$

$$Op(x, y, z, t) = +r1_x\ r1_x + r1_y\ r1_y + r1_z\ r1_z \quad (82)$$

Auxiliary analysis variable equation

$$r2_x = Op(x, y, z, t) \quad (83)$$

$$Op(x, y, z, t) = +x - \frac{2}{1 + mu} rorbit_dynamic \cos (omegaorbit\ t) \quad (84)$$

Auxiliary analysis variable equation

$$r2_y = Op(x, y, z, t) \quad (85)$$

$$Op(x, y, z, t) = +y - \frac{2}{1 + mu} rorbit_dynamic \sin (omegaorbit\ t) \quad (86)$$

Auxiliary analysis variable equation

$$r2_z = Op(x, y, z, t) \quad (87)$$

$$Op(x, y, z, t) = +z \quad (88)$$

Auxiliary analysis variable equation

$$sq_r2 = Op(x, y, z, t) \quad (89)$$

$$Op(x, y, z, t) = +r2_x\ r2_x + r2_y\ r2_y + r2_z\ r2_z \quad (90)$$

Auxiliary analysis variable equation

$$Tbar = Op(x, y, z, t) \quad (91)$$

$$Op(x, y, z, t) = - \left(sq_r1\ e^{-\frac{(\sqrt{sq_r1} - rdonut)^2}{p_sigma^2}} + mu\ (Nstar \right. \\ \left. - 1)\ sq_r2\ e^{-\frac{(\sqrt{sq_r2} - rdonut)^2}{p_sigma^2}} \right) \frac{\frac{mass}{1+mu}}{massfactor} \frac{1}{(6.28318530717959\ p_sigma)^{\left(\frac{3}{2}\right)}} \quad (92)$$

Auxiliary analysis variable equation

$$omegaorbit = Op(x, y, z, t) \quad (93)$$

$$\begin{aligned}
Op(x, y, z, t) = & \quad (94) \\
+ p_Omega \sqrt{2 \frac{mass}{(2 \ rorbit_dynamic)^3}} & \left(\max\{0, \min\{1, 1 - \max\{0, \min\{1, t - torbit\}\}\}\} \max\{0, \min\{1, \max\{0, \min\{1, t - torbit\}\}\}\} \right. \\
& \times 4 \ (2 \ torbit - t) \ \frac{t}{(2 \ torbit)^2} + \max\{0, \min\{1, \max\{0, \min\{1, t - torbit\}\}\}\} \left. \right)
\end{aligned}$$

Auxiliary analysis variable equation

$$rorbit_dynamic = (x, y, z, t) \quad (95)$$

$$\begin{aligned}
(x, y, z, t) = & + \max\{0, \min\{1, \max\{0, \min\{1, ((t - tini) - tslow) - torbit\}\}\}\} (96) rbit \\
& + \max\{0, \min\{1, 1 - \max\{0, \min\{1, ((t - tini) - tslow) - torbit\}\}\}\} \max\{0, \min\{1, \max\{0, \min\{1, (t - tini) - torbit\}\}\}\} \left(rstart \right. \\
& + \frac{6}{tslow^3} (rorbit - rstart) \left(\left(-\frac{((t - tini) - torbit)^3}{3} \right) \right. \\
& \left. \left. + tslow \frac{((t - tini) - torbit)^2}{2} \right) \right) \\
& + rstart \max\{0, \min\{1, 1 - \max\{0, \min\{1, (t - tini) - torbit\}\}\}\}
\end{aligned}$$

Auxiliary analysis variable equation

$$Vtx = Op(x, y, z, t) \quad (97)$$

$$\begin{aligned}
Op(x, y, z, t) = & + Phid_z \ Phid_x \ \partial_z Psisf + Phid_y \ Phid_x \ \partial_y Psisf \\
& + Phid_x^2 \ \partial_x Psisf + Psisf^2 \ \partial_x Psisf \\
& - Phid_z \ Psisf \ \partial_z Phid_x - Phid_y \ Psisf \ \partial_y Phid_x \\
& - Phid_x \ Psisf \ \partial_x Phid_x - d2tphi \ Phid_x \ Psisf
\end{aligned} \quad (98)$$

Auxiliary analysis variable equation

$$Vty = Op(x, y, z, t) \quad (99)$$

$$\begin{aligned}
Op(x, y, z, t) = & +Phid_z \ Phid_y \ \partial_z Psisf + Phid_y^2 \ \partial_y Psisf \\
& + Psisf^2 \ \partial_y Psisf + Phid_y \ Phid_x \ \partial_x Psisf \\
& - Phid_x \ Psisf \ \partial_y Phid_x - Phid_z \ Psisf \ \partial_z Phid_y \\
& - Phid_y \ Psisf \ \partial_y Phid_y - dztphi \ Phid_y \ Psisf
\end{aligned} \tag{100}$$

Auxiliary analysis variable equation

$$Vtz = Op(x, y, z, t) \tag{101}$$

$$\begin{aligned}
Op(x, y, z, t) = & +Phid_z^2 \ \partial_z Psisf + Psisf^2 \ \partial_z Psisf \\
& + Phid_z \ Phid_y \ \partial_y Psisf + Phid_z \ Phid_x \ \partial_x Psisf \\
& - Phid_x \ Psisf \ \partial_z Phid_x - Phid_y \ Psisf \ \partial_z Phid_y \\
& - Phid_z \ Psisf \ \partial_z Phid_z - dztphi \ Phid_z \ Psisf
\end{aligned} \tag{102}$$

Auxiliary analysis variable equation

$$Vxy = Op(x, y, z, t) \tag{103}$$

$$\begin{aligned}
Op(x, y, z, t) = & +Phid_z \ Phid_y \ \partial_z Phid_x + Phid_y^2 \ \partial_y Phid_x \\
& - Phid_x^2 \ \partial_y Phid_x + Phid_y \ Phid_x \ \partial_x Phid_x \\
& - Phid_y \ Psisf \ \partial_x Psisf - Phid_z \ Phid_x \ \partial_z Phid_y \\
& - Phid_y \ Phid_x \ \partial_y Phid_y + Phid_x \ Psisf \ \partial_y Psisf
\end{aligned} \tag{104}$$

Auxiliary analysis variable equation

$$Vxz = Op(x, y, z, t) \tag{105}$$

$$\begin{aligned}
Op(x, y, z, t) = & +Phid_z^2 \ \partial_z Phid_x - Phid_x^2 \ \partial_z Phid_x \\
& + Phid_z \ Phid_y \ \partial_y Phid_x + Phid_z \ Phid_x \ \partial_x Phid_x \\
& - Phid_z \ Psisf \ \partial_x Psisf - Phid_y \ Phid_x \ \partial_z Phid_y \\
& - Phid_z \ Phid_x \ \partial_z Phid_z + Phid_x \ Psisf \ \partial_z Psisf
\end{aligned} \tag{106}$$

Auxiliary analysis variable equation

$$Vyz = Op(x, y, z, t) \quad (107)$$

$$\begin{aligned} Op(x, y, z, t) = & +Phid_z \ Phid_x \ \partial_y Phid_x + Phid_z^2 \ \partial_z Phid_y \\ & - Phid_y^2 \ \partial_z Phid_y + Phid_z \ Phid_y \ \partial_y Phid_y \\ & - Phid_z \ Psisf \ \partial_y Psisf - Phid_y \ Phid_x \ \partial_z Phid_x \\ & - Phid_z \ Phid_y \ \partial_z Phid_z + Phid_y \ Psisf \ \partial_z Psisf \end{aligned} \quad (108)$$

Boundary conditions

Segments: main

Type: extrapolation
Axis: All
Side: All

Fields: *phi*, *Phid_x*, *Phid_y*, *Phid_z*, *Psisf*, *phi2*, *dphi2*, *Psi*

Boundary precedences

1. x-Lower
2. x-Upper
3. y-Lower
4. y-Upper
5. z-Lower
6. z-Upper

Finalization Conditions

The condition is:

$$t \geq tend \quad (109)$$